

CS224N: NLP with Deep Learning

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

Contents

Course map	2
Prerequisites	5
Notation and shapes	8
Glossary	11
Cheatsheet	17
Crash course	20

Course map

Source: official schedule table at <http://web.stanford.edu/class/cs224n/>, fetched 2026-10-06. Disclaimer carried verbatim: the schedule is tentative and subject to change. U01-U08 were built by the first builder, U09-U15 by the second builder. Every session below is mapped to a unit. No session is orphaned.

#	Date (2026)	Session	Unit	Claim class	Note
S01	Tue Jan 6	History of NLP	U01	OFFICIAL-SOURCE (SRC-02)	A1 out: Introduction to word vectors
S02	Thu Jan 8	Word Vectors	U02	OFFICIAL-SOURCE	Readings: word2vec, negative sampling, GloVe papers (titles only)
S03	Fri Jan 9	Python Review Session	U01/U15	OFFICIAL-SOURCE	Bridge session, maps to P02 remediation
S04	Tue Jan 13	Backpropagation and Neural Network Basics	U03	OFFICIAL-SOURCE	A2 out (neural nets, tensor derivatives, dependency parsing), A1 due
S05	Thu Jan 15	Language Models and RNNs	U04	OFFICIAL-SOURCE	Readings: vanishing gradient papers, Attention Is All You Need (titles)
S06	Fri Jan 16	PyTorch Tutorial Session	U03/U04	OFFICIAL-SOURCE	Maps to P12 remediation
S07	Tue Jan 20	Transformers	U05	OFFICIAL-SOURCE	Readings: Attention paper, Illustrated Transformer, LayerNorm (titles)
S08	Thu Jan 22	Final Projects: Custom and Default, Practical Tips	U15	OFFICIAL-SOURCE	A3 out (self-attention and transformers), A2 due
S09	Tue Jan 27	Pretraining (Scaling, Systems, Data)	U06	OFFICIAL-SOURCE	Readings: BERT, ELMo, Llama 3 (titles)
S10	Thu Jan 29	Post-training (RI HE SET) ³	U07	OFFICIAL-SOURCE	Readings: InstructGPT

Mismatches flagged

- Dependency parsing: A2's official title includes "dependency parsing",

but no scheduled lecture carries that name. U03-C07 (parsing state/actions) and U03-C08 (supervised objective) are mapped to A2 plus the W2 neural-net sessions. Claim class: OFFICIAL-SOURCE for the A2 title, REQUESTED-BRANCH for lecture-level attribution.

- "Efficient Adaptation (Prompting + PEFT)" reads as one session, the

unit U08 splits prompting (C01-C06) from adaptation (C07-C12). The split is a teaching choice, not an official division.

- S12 "Agents, Tool Use, and RAG" and the tutorial sessions are

built in U09 (agents) and U15 (tutorials). They were mapped early so no session was orphaned.

Prerequisites

Shared modules P01-P24 live at `../shared/prerequisites/` and are linked, never rebuilt. Each unit lists its parent prerequisites from the course prompt, each lesson carries a local remediation block for the misses that matter most. Take the diagnostic before U01. Score each item 0 (miss), 1 (partial), or 2 (full). Key: `keys/diagnostic_key.md`.

Unit prerequisite map

Unit	Parent prereqs	Bridge content	Remediation pointer in lesson
U01	P02, P06, P10, P13	Python and scientific software, probability, ML foundations and evaluation, language and sequence modelling	Lesson U01 block: n-gram counting and split discipline
U02	P03, P05, P08	Vectors and linear maps, calculus, information theory	Lesson U02 block: dot product, softmax, cross-entropy
U03	P05, P11, P12	Calculus, neural nets and autodiff, PyTorch and stability	Lesson U03 block: computational graph and chain rule
U04	P11, P13	Neural nets and autodiff, language and sequence modelling	Lesson U04 block: teacher forcing and perplexity
U05	P12, P14	PyTorch and stability, transformer mechanics	Lesson U05 block: Q/K/V shapes and causal mask
U06	P10, P14, P15	ML foundations, transformer mechanics, hardware	Lesson U06 block: FLOPs counting and tokens budget
U07	P08, P14, P17	Information theory, transformer mechanics, RL	Lesson U07 block: KL, Bradley-Terry, policy gradient
U08	P09, P14	Optimization, transformer mechanics	Lesson U08 block: LoRA rank and parameter counts
U09	P19, P20, P21	Retrieval and information access, tools, APIs, and agent state, security, privacy, and safety	Lesson U09 block: schema validity, recall at k, split rate
U10	P07, P10, P22	Statistical estimation and uncertainty, ML foundations and evaluation, experimental method and research	Lesson U10 block: Wilson interval, contamination arithmetic, sample size

Diagnostic (closed book, 20 minutes)

D1. Encode "€" (U+20AC) in UTF-8 by hand. State the byte count. D2. With a fair coin, compute the cross-entropy between the true distribution and the model prediction (0.5, 0.5). State units. D3. Write the dot product of $u=(1,2)$ and $v=(3,-1)$. State the angle sign: positive, zero, or negative. D4. Differentiate $f(x)=\log(1+\exp(x))$ with respect to x . Name the result. D5. Softmax of (1000, 1001, 1002). Compute without a calculator, by shifting first. State the largest entry. D6. A batch of 4 sequences, each length 7, vocab 50. State the shape of the logit tensor and of the target tensor for next-token loss. D7. Define perplexity in one sentence and give the perplexity of a model that assigns probability 0.25 to every correct next token. D8. Sketch the Q/K/V shapes for one attention head: sequence length 6, head dim 8, batch 2. Include the score matrix shape. D9. Estimate the FLOPs of one dense layer, input dim 1024, output dim 4096, batch 32. Count multiply-add as 2 FLOPs. D10. A reward model prefers response A over B. Write the Bradley-Terry probability that A wins, in terms of scalar scores r_A and r_B .

Scoring: 16-20 , proceed. 10-15 , read the flagged remediation blocks. Below 10 , work P01-P13 first. The diagnostic is a placement tool, not a mastery claim.

Notation and shapes

Symbols keep one meaning across all eight units. Shapes are given as (batch, sequence, feature) unless stated. All tensors are float32 on CPU in the executed scripts unless a script states otherwise.

Scalars

Symbol	Meaning	Unit
V	vocabulary size	count
d	embedding or model dimension	count
d_k, d_h	attention head dimension	count
h	number of attention heads	count
T, n	sequence length	tokens
B	batch size	count
L	number of layers	count
P	parameter count	count
D	training token count	tokens
C	training compute	FLOPs
eta, lr	learning rate	1/step
beta	DPO or KL coefficient	scalar
gamma	discount factor (RL)	[0, 1]
ppl	perplexity	dimensionless

Vectors and matrices

Symbol	Shape	Meaning
x_i	$(d,)$	input embedding of token i
$e(w)$	$(d,)$	embedding row of word w
E	(V, d)	embedding matrix
W	$(d_{\text{out}}, d_{\text{in}})$	weight matrix
Q, K, V	(B, n, d_k) per head	query, key, value
A	(n, n) per head	attention weights, rows sum to 1
z	$(V,)$	logits
$p, \hat{y}$	$(V,)$	predicted distribution
y	$()$ int or $(V,)$ one-hot	target token or distribution
h_t	$(d,)$	recurrent hidden state at step t
s_t	$(n,)$	parser stack contents at step t

Operators

- $\text{softmax}(z)_i = \exp(z_i - \max z) / \sum_j \exp(z_j - \max z)$.
- cross-entropy $H(p, q) = -\sum_i p_i \log q_i$. Log is natural log, units

are nats. Perplexity = $\exp(\text{cross-entropy})$.

- $KL(p \parallel q) = \sum_i p_i \log(p_i / q_i)$, nats.
- $\cos(u, v) = u \cdot v / (||u|| ||v||)$.
- $\text{sigma}(x) = 1 / (1 + \exp(-x))$.

Index conventions

- i, j index sequence positions, k indexes vocabulary or heads.
- w_t is token at position t , $w_{\{<t\}}$ is the prefix before t .
- Superscripts in parentheses, e.g. $x^{(l)}$, index layers.
- A colon slice $a[i:j]$ follows Python rules: includes i , excludes j .

Shape contracts (checked in every implementation)

- Embedding lookup: (B, n) ids $\rightarrow (B, n, d)$.
- Attention scores: $(B, h, n, d_h) \times (B, h, d_h, n) \rightarrow (B, h, n, n)$.
- Next-token logits: $(B, n, d) \rightarrow (B, n, V)$. Loss drops the last position: targets (B, n) use $w_2..w_{n+1}$ against logits at $1..n$.
- Masked LM: loss over masked positions only, shape (num_masked, V) .

Glossary

One meaning per term. First-use definitions in lessons match these.

- ambiguity: one input admits two or more valid interpretations.
- analogy: $a:b :: c:d$ tested by vector arithmetic on embeddings.
- attention: weighted mix of values by query-key similarity.
- autoregressive: each token predicted from the prefix only.
- batch: one set of examples processed together in an update.
- beam search: decoding that keeps the top-k partial hypotheses.
- Bradley-Terry: model of pairwise preference from scalar scores.
- CBOW: continuous bag of words, predicts a center word from context.
- chunking: splitting documents into retrieval-sized pieces.
- cooccurrence: two words appearing in one context window.
- cosine similarity: dot product divided by both norms, in $[-1, 1]$.
- cross-entropy: negative log-likelihood of the true distribution.
- decoding: turning model scores into a token sequence.
- distributional hypothesis: words in similar contexts have similar meanings.
- DPO: direct preference optimization, preference loss without a separate reward model or RL loop.
- embedding: a learned vector that stands for a discrete token.
- encoder: a block that reads the full input, no causal mask.
- decoder: a block that generates left to right under a causal mask.
- exposure bias: train sees gold prefixes, test sees model prefixes.
- feedforward: position-wise two-layer MLP inside a transformer block.

- GloVe: count-based embedding method on a cooccurrence matrix.
- gradient clipping: rescale a gradient whose norm passes a cap.
- greedy decoding: always pick the highest-probability token.
- hierarchical softmax: softmax factorized over a binary tree.
- instruction tuning: supervised fine-tuning on instruction-response pairs.
- intrinsic evaluation: embedding quality judged by word tasks, not by downstream models.
- KL divergence: expected log ratio of p to q under p , in nats.
- KV cache: stored keys and values that avoid recompute in decoding.
- LoRA: low-rank additive update to a frozen weight matrix.
- masked LM: predict masked tokens from both directions of context.
- multi-head attention: parallel attention heads with separate projections, concatenated at the output.
- negative sampling: binary classification against sampled noise words.
- one-hot: vector with a single 1 and zeros elsewhere.
- padding: dummy tokens that fill a batch to one length.
- PEFT: parameter-efficient fine-tuning, update of a small subset.
- perplexity: $\exp$ of mean negative log-likelihood, in "choices".
- PMI: pointwise mutual information, log of observed over expected joint probability.
- policy: the model viewed as a mapping from context to action distribution (RL framing).
- polysemy: one word form with several distinct senses.

- post-training: training after pretraining: SFT, preferences, RL.
- pretraining: first-stage training on a broad corpus.
- PPO: proximal policy optimization, clipped policy-gradient method.
- prompt sensitivity: output changes under paraphrase of the prompt.
- reward hacking: a policy that raises the proxy reward without raising the true objective.
- reward model: a scorer trained on human preference pairs.
- RLHF: reinforcement learning from human feedback.
- scaling law: fitted relation of loss to parameters, data, compute.
- self-consistency: majority vote over sampled reasoning paths.
- SFT: supervised fine-tuning on demonstration data.
- skip-gram: predict context words from a center word.
- softmax: normalized exponential map from scores to probabilities.
- teacher forcing: train the next-token model on gold prefixes.
- token: the atomic text unit the model reads and writes.
- top-k sampling: sample from the k most likely tokens only.
- top-p sampling: sample from the smallest set with total mass p.
- vanishing gradient: gradient norm shrinks across long dependencies, so distant inputs stop moving the parameters.
- window: the fixed span of context around a center word.
- zero-shot: task solved with no examples in the prompt.
- acceptance criterion: pre-registered pass/fail contract for a decision, written before measuring.
- aux loss: extra training loss that balances expert routing.

- BPE: byte-pair encoding, merge frequent adjacent pairs.
- Brier score: mean squared error of stated probabilities.
- canary: planted string used to test memorization.
- causal intervention: change a component, measure the behavior change.
- citation: pointer from a claim to a supporting span.
- claim class: honesty label for a statement (official, requested-branch, restricted, edition tag).
- code point: Unicode number for a character.
- concept direction: vector in activation space that separates labeled examples.
- contamination: test items present in training data.
- context: the working token window of the current run.
- contribution: the built column of the project ledger.
- coverage matrix: benchmark tasks vs deployment capabilities.
- defense rubric: claim, evidence, limit, transfer, four points.
- disparity ratio: min group rate over max group rate.
- ECE: expected calibration error, mean $|\text{confidence} - \text{accuracy}|$.
- error attribution: labeling a failure R (retriever), T (tool), or P (policy).
- fertility: tokens per word in a language.
- guest card: title, source, inspection status, claims (none until inspected).
- hybrid retrieval: sparse plus dense scores fused by weight.

- impact report: scored checklist before shipping a system.
- kappa: chance-corrected agreement between judges.
- memory: stored state across turns (verbatim, summary, anchors).
- misinformation: fluent false claims at scale.
- modality expert: specialist sub-network per input sense.
- open question: dated unknown with a closing criterion.
- permission scope: the set of tools a run may call.
- RAG: retrieve top-k chunks, prepend, generate.
- ReAct: interleaved Thought / Action / Observation loop.
- reading card: title, claim, read status for a paper.
- red team: adversarial probing of a system's harms.
- rerank: heavy rescoring of a small retrieved pool.
- reproducibility receipt: seeds, versions, data, commands.
- seed sweep: run k seeds, report mean and std.
- slice: subgroup score with its delta from the mean.
- speculative decoding: draft with a small model, verify with the big one, same output distribution.
- split rate: fraction of answer spans cut by chunk boundaries.
- steering: add alpha times a direction to activations.
- step cap: max agent loop steps, the fire escape.
- tool schema: name, argument schema, return schema, side-effect class.
- trace: the model's step-by-step text, a story not a proof.
- transfer gap: source accuracy minus target accuracy.
- translationese: simplified language of translated text.

- UTF-8: variable-length byte encoding of code points.
- verifiable reward: reward a program checks (tests, exact match).
- Wilson interval: confidence interval for a proportion.

Cheatsheet

One page per area. Numbers are toy-scale from the unit scripts.

Core formulas

- Perplexity: $\exp(\text{mean negative log-likelihood})$, per tokenizer.
- Attention: $\text{softmax}(QK^T / \sqrt{d_k}) V$, mask before softmax.
- LoRA: $dW = BA$, rank $\leq r$, params $2dr$ per matrix.
- Majority vote: $\sum C(n,k) p^k (1-p)^{(n-k)}$, $k > n/2$.
- Wilson: point $\pm z \sqrt{p(1-p)/n + z^2/4n^2}$, over $1 + z^2/n$.
- Kappa: $(p_o - p_e) / (1 - p_e)$.
- Brier: $\text{mean}((f - y)^2)$. ECE: $\text{mean}|\text{conf} - \text{correct}|$.
- Contamination: $\text{reported} = L s_m + (1 - L) s_c$.
- Split rate: $\max(0, s - o) / (c - o)$.
- Chunks: $1 + \text{ceil}((\text{doc} - c) / (c - o))$.
- Fusion: $w \text{ dense} + (1 - w) \text{ sparse}$.
- Disparity: $\text{min rate} / \text{max rate}$.
- Spec decode E: $(1 - a^{(g+1)}) / (1 - a)$.
- Aux loss: $\text{mean}((\text{load} / \text{mean})^2)$, 1.0 is balanced.

Key numbers

- LoRA $d=4096$ $r=8$: 65536/matrix, 0.060% of 7B.
- Vote $p=0.6$: 0.6000, 0.6826, 0.7535, 0.8256 ($n=1,5,11,21$).
- Vote $p=0.4$: 0.3174 ($n=5$), 0.2465 ($n=11$): do not vote.
- 78/100: [0.689, 0.850]. 780/1000: [0.753, 0.805].
- $n = 2401$ for margin 0.02 at 95%.

- Contamination: 0.75 reported, 0.70 clean.
- Kappa 0.551, Brier 0.17 on the toys.
- RAG recall@k: 0.000, 0.333, 0.667, 1.000.
- Chunk 200/50: 7 chunks, split 0.00. No overlap: 5, 0.15.
- Fertility: 1.3, 1.6, 2.1, 2.8. Cost ratios: 1.00-2.42x.
- BPE toy: 6 merges, 11 chars to 2 pieces.
- Disparity 0.667. Intervention effect 1.5. ECE 0.425.
- Late fusion 5.03e7 params. Routing aux 1.04. Image share 0.333.
- Tiny GPT-2: 44,928 params. Seeds: 0.8045 +/- 0.0203.
- ReAct toy: 7 steps, 92 tokens. Spec decode: 2.94/pass.

Decision rules

- Conditioning is not learning: prompts steer, they do not train.
- Report the interval, never the bare number.
- Attribute before fixing: R/T/P, biggest class first.
- Spread concave gains: 10x10 beats 1x100, except thresholds.
- Check p before voting: below 0.5 voting hurts.
- Cap the agent loop: valid answer or step cap, both exits.
- Enforce permissions in code, not in the prompt.
- Calibrate the judge: kappa and Brier before trust.
- Quarantine leaks: rescore on the clean set.
- Read the slice, not the mean.
- Pre-register the acceptance criterion.
- Pin the model revision. Keep the receipt (seeds, versions).
- Cut scope, never the control group.

- Guest content: title only until inspected. Demote on doubt.
- Traces are evidence, not proof: run the corruption test.

Claim classes

- OFFICIAL-SOURCE: inspected artifact (schedule S01-S20, SRC-01-06).
- REQUESTED-BRANCH: prompt/inventory request, toy-taught, PLANNED /
SOURCE ATTRIBUTION PENDING.
- RESTRICTED-UNVIEWED: exists behind login, not viewed.
- EDITION-2024: 2024 public videos, tagged.

Crash course

The whole course in one pass. Each unit: the idea, the number, the trap. Numbers come from the unit compute scripts, all toy-scale and labeled as such.

U01 , NLP history, tasks, data

Idea: language is ambiguous, tasks need formulation (loss, metric, data) before modeling. Number: the course starts from counting. Trap: judging a task by its name.

U02 , word vectors

Idea: the distributional hypothesis, skip-gram/CBOW, negative sampling, GloVe, PMI. Number: analogy cosines near 1.0 on the toy. Trap: reading cosine as probability.

U03 , neural fundamentals and parsing

Idea: the computational graph, chain rule, embeddings as parameters, shift/reduce parsing. Number: shift by max before softmax. Trap: unshifted softmax NaNs.

U04 , language models and recurrence

Idea: teacher forcing, perplexity, BPTT, vanishing gradients, LSTM/GRU, decoding. Number: perplexity is per-token, compare only within one tokenizer. Trap: teacher forcing vs free running.

U05 , attention and transformers

Idea: Q/K/V, $1/\sqrt{d_k}$, MHA, masks, residuals, positions, FFN, KV cache. Number: the n^2 attention bill. Trap: the causal mask goes before softmax.

U06 , pretraining, scaling, systems, data

Idea: masked and autoregressive objectives, corpus curation, scaling laws, leakage, transfer, model cards. Number: doubling compute cuts loss about 3.4 percent on the toy curve. Trap: scaling params without data.

U07 , post-training and preferences

Idea: SFT, reward models, Bradley-Terry, RLHF/PPO, DPO, KL leash, reward hacking, annotation bias. Number: reward gap 0.9 gives P 0.711. Trap: optimizing the proxy and calling it aligned.

U08 , prompting and efficient adaptation

Idea: zero/few-shot, in-context learning, prompt sensitivity, chain-of-thought (useful, not faithful), self-consistency, adapters, LoRA, forgetting, validation. Number: LoRA trains 0.060 percent of params. Majority of 5 at p 0.6 gives 0.683. Trap: conditioning is not learning.

U09 , tools, agents, and retrieval

Idea: ReAct (Thought/Action/Observation), tool schemas, execution feedback, RAG, retriever/generator split, chunking, hybrid/rerank, citations, memory vs context, permissions, stopping, attribution. Number: 7 steps, 92 tokens. Recall at k 0.000/0.333/0.667/1.000. split rate 0.15 to 0.00 with overlap 50. Trap: the loop never stops on its own.

U10 , benchmarking and research methodology

Idea: benchmark families, coverage, judge calibration, contamination, uncertainty, scoring formats, replication, slices, bias, acceptance criteria, project choice, contribution accounting. Number: 78/100 is [0.689, 0.850]. 40 leaked items inflate 0.70 to 0.75. kappa 0.551. Trap: the bare number, the hidden slice.

U11 , reasoning and test-time compute

Idea: reasoning supervision, verifiable rewards, outcome vs process verification, repeated sampling, self-consistency, compute allocation, speculative decoding, eval bias, failure analysis, cost-quality, trace limits. Number: vote curve 0.600 to 0.826. spec decode 2.94 tokens per pass. 10x10 beats 1x100 (3.94 vs 0.99). Trap: voting below p 0.5, trusting traces.

U12 , tokenization and multilinguality

Idea: Unicode, BPE, vocab sharing, rare words, transfer, multilingual space, cost inequality, low-resource data, parity, morphology, code switching, fairness bets. Number: fertility 1.3 to 2.8. cost ratios to 2.42x. BPE 6 merges on the toy. Trap: the same model at different prices.

U13 , interpretability and social impacts

Idea: concept probes, behavioral vs mechanistic evidence, causal interventions, vocabulary limits, steering, attribution ladder, misinformation, bias, privacy, misuse, uncertainty, impact reports. Number: disparity ratio 0.667. intervention effect 1.5. ECE 0.425. extraction 0.185. Trap: "truth direction" headlines.

U14 , multimodality and guest research

Idea: patch tokens, early/late fusion, mixed streams, joint generation, diffusion/AR mix, modality experts, multimodal retrieval, instruction tuning, reward eval, guest cards, claim classes, gap ledger. Number: late fusion 5.03e7 params vs 0. routing aux 1.04. image share 0.333. Trap: input format is not multimodality. hearsay is not a claim.

U15 , project tutorials and open questions

Idea: Python, PyTorch, Hugging Face, tiny GPT-2, downstream tasks, the four gates, scoping, originality, reproducibility, limits, open questions, defense. Number: 44,928 params. seeds mean 0.8045, std 0.0203. Trap: toy numbers cited as findings, one seed, the cut control group.

The course in five lines

Formulate before modeling. Condition is not learning. Report the interval, not the number. Attribute before fixing. Every claim carries its evidence and its limit.